

Experimental Configurations

Fixed τ configuration:

$\tau = 100$ d for every curve,
isolating the method from
any spread of dynamics

Varying τ configuration:

τ drawn from the quasar mass
distribution (Sun+23) via a
mass–timescale relation (Burke+21)

Simulation

Dense DRW realization:

sampled with EZTAO (GP, DRW kernel)
on a fine grid, true signal $\rightarrow (t_{\text{true}}, y_{\text{true}})$

Cadence sampling & gaps:

draw epochs, remove contiguous intervals

Photometric noise:

add heteroscedastic Gaussian noise at the
LSST single-visit precision 0.01 mag
 $\rightarrow (t_{\text{obs}}, y_{\text{obs}}, \sigma_{\text{obs}})$

dense realization kept as
ground truth alongside the
sparse noisy observations

Preprocessing

Padding & masks:

pad to T_{max} , validity mask

Train/validation split:

80% train, 20% validation

Training Loop

Neural stochastic flow & bridge:

affine-coupling flow and bridge
Kiyohara+25, *trained
separately per configuration*

flow $p_{\theta}(x_t | x_s, \Delta t)$
and bridge
 $q_{\xi}(x_j | x_i, x_k)$

Triplet training:

data likelihood + λ flow-consistency,
optimized with Adam over (θ, ξ)

triplets (i, j, k) , $i < j < k$;
one-step + bridge
vs two-step flow

Trained model

one instance per configuration

Evaluation

Evaluation vs ground truth:

on the validation split, compared
with the latent signal and GP oracle

coverage: calibration,
RMSE: accuracy,
predictive LL: both